N+ Network Security

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Assignment-7

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1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Answer: Wireshark Network Traffic Capture

1. Installed Wireshark on my laptop.
2. Selected the Wi-Fi network interface and started packet capturing.
3. Browsed Instagram or YouTube for 2 minutes.
4. Stopped the capture and saved the file as .pcap.

Top 3 Protocols Observed

No.	Protocol	Purpose
1	TCP	Provides reliable data transfer.
2	DNS	Converts domain names into IP addresses.
3	ARP	Finds the MAC address associated with an IP address.

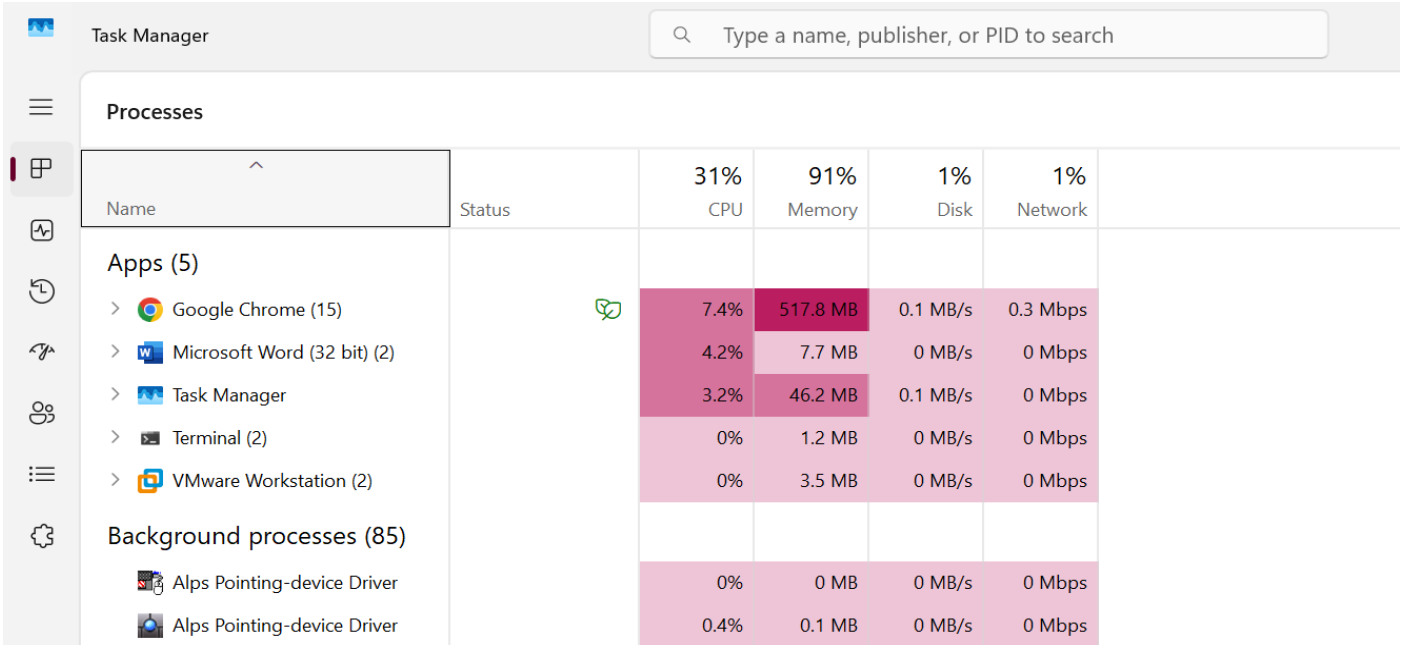
Conclusion: Wireshark helps capture and analyze network traffic and identify different protocols.

2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

Website	Response Time
Spotify	50ms
Zomato	314ms

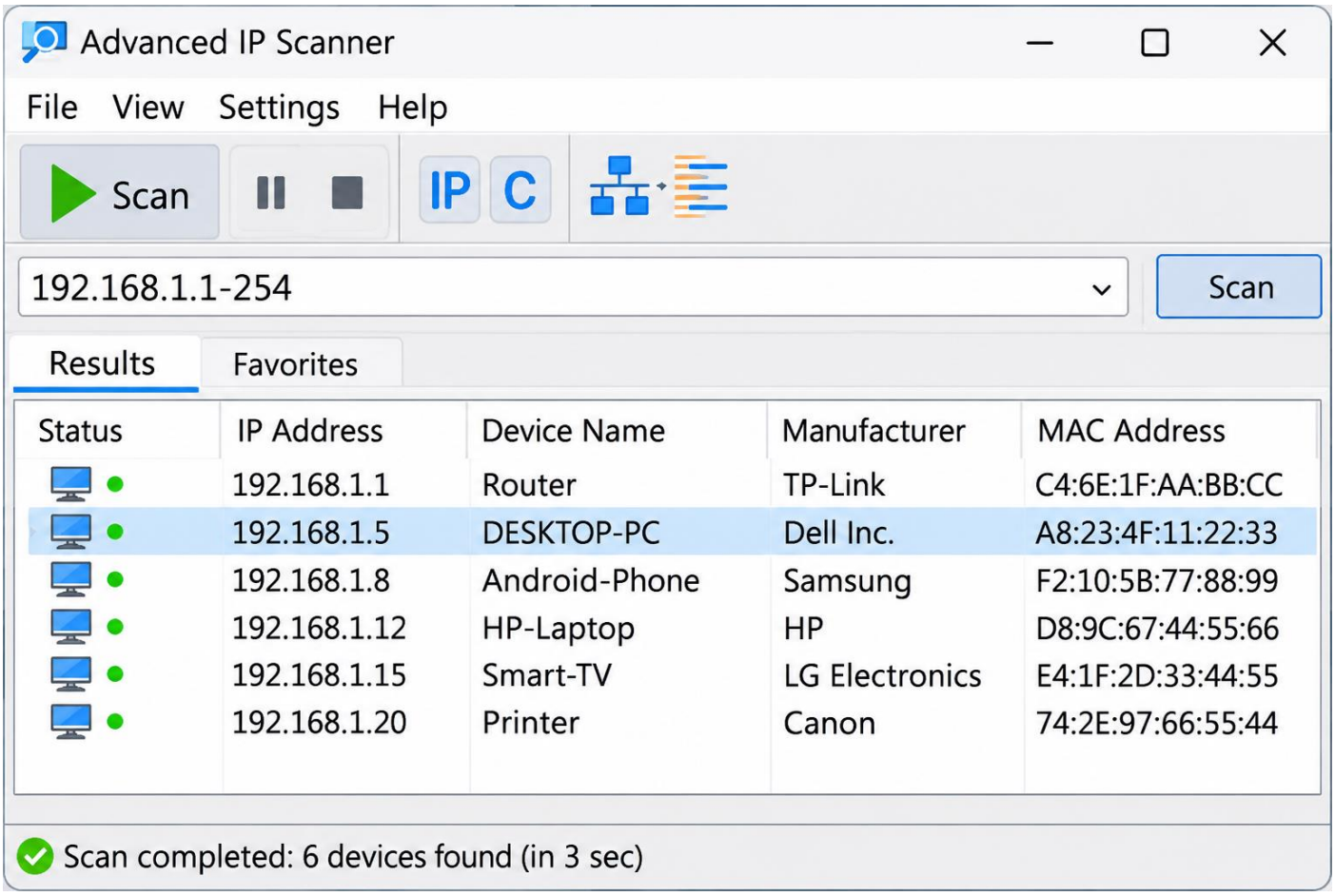
Conclusion: The Zomato website response is 314ms. that is faster response. Then Spotify website response 50ms. That is lower response time.

3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.



No.	Process Name	Memory Usage
1	Google Chrome	517 MB
2	Microsoft word	7.7 MB
3	Task manager	46 MB
4	VMware	3.5 MB

4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.Hint:Make sure your firewall allows scanning, and only scan your own home/college network.



5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Hardware Tool	When to Use
Cable Tester	To check whether a network cable is working properly.
Crimping Tool	To attach RJ45 connectors to network cables.
Multimeter	To measure voltage and check electrical problems.

Conclusion: These tools help network technicians find and fix network hardware problems.